Statements equivalent to the proof of the Collatz conjecture, useful sequences, and a contradiction leading to the disproof of the conjecture

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Abstract

We look at the Collatz conjecture by using finite recurrences. We start with five types of numbers that follow the conjecture. We show that all of these types end up as two final forms. Both final forms are controlled by a sequence of exponents written as (p_i). Because of this the conjecture is the same as saying that every odd number has the first final form of the article. We prove that the length N of this expression can be different. We explain how we can always choose the possible N. We show many groups of odd numbers that have the first final form. One group is made using the Fermat–Euler theorem. We also add a part with a way to calculate numbers like $(2^{2M}-1)/3$ without dividing. In the end we assume that the numbers $4B+1$ and $4B-1$ both follow the conjecture. When we compare their forms we find a problem. This leads us to say that the conjecture is not true!

Keywords: Collatz conjecture, Syracuse, sequence, process, recurrence, even, odd, expression, natural numbers, solution, form, separate sets, equivalent proof, contradiction, disproof.

1 Introduction

Many works still try to explain the meanings and the consequences of the Collatz conjecture [1], whereas other works propose new approaches in studying this problem [2]. And thus, we have the right to believe that the field of mathematics has all the tools to find solution of arithmetic problems such as the Collatz conjecture. We carry on consequently in our previous efforts in proposing new results [3] in order to better explain the aspects of this conjecture and in order to collaborate with all the lovers of mathematics who dream of finding a solution for the Collatz conjecture.

In our previous work [3], we concluded for $x_n \neq 0$ that all the natural numbers x_n which respect the Collatz conjecture can be expressed in at least one of the following five forms.

The first form:

$$x_n = 2^M \qquad \forall M \in \mathbb{N} \setminus \{0\}.$$

The second form:

$$x_n = \frac{2^{2\times M}-1}{3} \qquad \forall M \in \mathbb{N} \setminus \{0\},$$

since the recurrence relation demonstrated in [3] is only for $n \geq 2$ and this is the case for $n = 0$.

The third form:

$$3 \times x_n = 2^{2\times M+p_1} - 2^{p_1} \qquad \forall M \in \mathbb{N} \setminus \{0\},$$

since the recurrence relation demonstrated in [3] is only for $n \geq 2$ and this is the case for $n = 1$. In this expression p_1 can be any natural number with $p_1 \geq 1$.

The fourth form:

$$3^{\frac{n}{2}+1} \times x_n + 3^{\frac{n}{2}} = 2^{\sum_{i=0}^{\frac{n}{2}} p_i} - \sum_{j=1}^{\frac{n}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n}{2}} p_i} \right),$$

where n can be any even natural number with $n \geq 2$ and the numbers p_i can be any natural numbers which respect $p_i \geq 1 \ \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times M$, where M can be any strictly positive natural number.

The fifth form:

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right),$$

where n can be any odd natural number with $n \geq 2$ and the numbers p_i can be any natural numbers which respect $p_i \geq 1 \ \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times M$, where M can be any strictly positive natural number.

We also concluded in [3] that we can prove that the Collatz conjecture is always true if we prove that all the natural numbers can be expressed in these five forms demonstrated above only.

We proposed that the artificial intelligence and many mathematics software can simplify this task of proving that all the natural numbers can be expressed in these five forms demonstrated above only. However, we work in this article without the help of computers in order to simplify even more the results of the previous article [3] that form the basis of the present work.

In this article we prove that all the numbers that respect the Collatz conjecture respect two final forms:

First final form: $3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right)$, with x_{2N} an odd natural number and N a natural number with $N \geq 2$;

Second final form: $3^M \times x_{2M-1} = 2^{\sum_{i=0}^{M} p_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} p_i} \right)$, with x_{2M-1} an even natural number and M a natural number with $M \geq 2$.

Here the numbers x_{2N} and x_{2M-1} are natural numbers which respect the Collatz conjecture and the numbers p_i can be any natural numbers which respect $p_i \geq 1 \ \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times A$, where A can be any strictly positive natural number.

We also prove that the numbers N and M are not unique in both expressions (Section 3). Precisely, if we have

$$3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right)$$

then we have also

$$3^{M+1} \times x_{2N} + 3^M = 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i} \right),$$

where M is another natural number with $M > N$ and the numbers q_k are natural numbers which respect $q_k \geq 1 \ \forall k \in \mathbb{N} \setminus \{0\}$ and $q_0 = 2 \times M'$, where M' is a strictly positive natural number, and we have $q_k = 2$ for $0 \leq k \leq M - N - 1$ and $p_{k-(M-N)} = q_k$ for any natural number k with $M - N \leq k \leq M$.

We can conclude that, for an odd number x_{2N} that respects the expression of the first final form, if N is the smallest natural number that satisfies $3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right)$, then all the other expressions of the first final form of this same number x_{2N} are with $M = N+O$, where O can be any strictly positive natural number, and we have also $3^{M+1} \times x_{2N} + 3^M = 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i}\right)$, the numbers q_k being natural numbers which respect $q_k \geq 1 \ \forall k \in \mathbb{N} \setminus \{0\}$ and $q_0 = 2 \times M'$, where M' is a strictly positive natural number, with $q_k = 2$ for $0 \leq k \leq M - N - 1$ and $p_{k-(M-N)} = q_k$ for any natural number k with $M - N \leq k \leq M$.

This means that we can build infinitely many other different expressions of the first final form for this same number x_{2N}. Consequently, the first final form expression of the number x_{2N} is not unique.

Furthermore, we prove that it is sufficient to prove that all odd natural numbers respect the first final form in order to prove the Collatz conjecture. We propose a method to find odd natural numbers that have a specific number N in their first final form (Section 7), and an annex gives a method to calculate all the possible values of the natural numbers $x_n = \frac{2^{2\times M}-1}{3}$ without any division, by means of a demonstrated sequence.

Finally, in Section 8 we consider that the sets of numbers $4 \times B + 1$ and $4 \times B - 1$ both respect the Collatz conjecture, each with its own first final form, and we compare the two expressions. This comparison yields a relation between the exponent sequences and then, in the author's argument, a contradiction which is interpreted as a disproof of the conjecture.

Remark (Notation)**.** Letters such as M, N, B, C, D, E, L are used locally: their meaning is fixed in the section where they appear (in particular in Sections 7 and 8).

2 The first form is included in the fifth form

Let n be an odd natural number with $n > 2$ and let M be a strictly positive natural number. We have

$$3^{\frac{n+1}{2}} \times 2^M = 3^{\frac{n+1}{2}-1} \times 3 \times 2^M = 3^{\frac{n+1}{2}-1} \times (2+1) \times 2^M = 3^{\frac{n+1}{2}-1} \times \left(2^{M+1} + 2^M\right) \quad (1)$$

Hence

$$3^{\frac{n+1}{2}} \times 2^M = 3^{\frac{n+1}{2}-1} \times \left(2^{M+1} + 2^{M+1} - 2^M\right) = 3^{\frac{n+1}{2}-1} \times \left(2^{M+2} - 2^M\right) \quad (2)$$

Consequently

$$3^{\frac{n+1}{2}} \times 2^M = 3^{\frac{n+1}{2}-1} \times 2^{M+2} - 3^{\frac{n+1}{2}-1} \times 2^M \quad (3)$$

We repeat the same method for

$$3^{\frac{n+1}{2}-1} \times 2^{M+2} \quad (4)$$

Hence we get

$$3^{\frac{n+1}{2}-1} \times 2^{M+2} = 3^{\frac{n+1}{2}-2} \times 2^{M+4} - 3^{\frac{n+1}{2}-2} \times 2^{M+2} \quad (5)$$

Consequently we have

$$3^{\frac{n+1}{2}} \times 2^M = 3^{\frac{n+1}{2}-2} \times 2^{M+4} - 3^{\frac{n+1}{2}-2} \times 2^{M+2} - 3^{\frac{n+1}{2}-1} \times 2^M \quad (6)$$

We repeat the same method for

$$3^{\frac{n+1}{2}-2} \times 2^{M+4} \quad (7)$$

Hence we get

$$3^{\frac{n+1}{2}-2} \times 2^{M+4} = 3^{\frac{n+1}{2}-3} \times 2^{M+6} - 3^{\frac{n+1}{2}-3} \times 2^{M+4} \quad (8)$$

Consequently we have

$$3^{\frac{n+1}{2}} \times 2^M = 3^{\frac{n+1}{2}-3} \times 2^{M+6} - 3^{\frac{n+1}{2}-3} \times 2^{M+4} - 3^{\frac{n+1}{2}-2} \times 2^{M+2} - 3^{\frac{n+1}{2}-1} \times 2^M \quad (9)$$

And thus we can guess the following recurrence relation:

$$3^{\frac{n+1}{2}} \times 2^M = 2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)} \tag{10}$$

Proof of the recurrence relation. For $n = 3$ we have

$$3^{\frac{n+1}{2}} \times 2^M = 9 \times 2^M \tag{11}$$

And also, for $n = 3$ we have

$$2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)} = 2^{M+4} - 3 \times 2^M - 2^{M+2} = 2^M \times \left(2^4 - 3 - 2^2\right) \tag{12}$$

Hence we have

$$2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)} = 9 \times 2^M \tag{13}$$

And thus, for $n = 3$ we respected that

$$3^{\frac{n+1}{2}} \times 2^M = 2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)}. \tag{14}$$

Now let us consider that this recurrence relation is correct up to the number n and let us prove it for $n + 2$, since n can only be an odd number. We have

$$3^{\frac{n+3}{2}} \times 2^M = 3^{\frac{n+1}{2}+1} \times 2^M = 3 \times 3^{\frac{n+1}{2}} \times 2^M \tag{15}$$

Consequently

$$3 \times \left(2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)}\right) = 3 \times 2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+3}{2}-i} \times 2^{M+2\times(i-1)} \tag{16}$$

And we have

$$3 \times 2^{M+n+1} = 2^{M+n+2} + 2^{M+n+1} = 2^{M+n+3} - 2^{M+n+1} \tag{17}$$

And thus we have

$$3\times \left(2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)}\right) = 2^{M+n+3} - 2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+3}{2}-i} \times 2^{M+2\times(i-1)} \tag{18}$$

And this is equivalent to

$$3 \times \left(2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)}\right) = 2^{M+n+3} - \sum_{i=1}^{\frac{n+3}{2}} 3^{\frac{n+3}{2}-i} \times 2^{M+2\times(i-1)} \tag{19}$$

Hence we proved that

$$3^{\frac{n+3}{2}} \times 2^M = 2^{M+n+3} - \sum_{i=1}^{\frac{n+3}{2}} 3^{\frac{n+3}{2}-i} \times 2^{M+2\times(i-1)} \tag{20}$$

and this means that our recurrence relation is correct also for $n+2$. We conclude that the recurrence relation $3^{\frac{n+1}{2}} \times 2^M = 2^{M+n+1} - \sum_{i=1}^{\frac{n+1}{2}} 3^{\frac{n+1}{2}-i} \times 2^{M+2\times(i-1)}$ is correct for any odd natural number n with $n > 2$ and for any strictly positive natural number M.

Also, by substituting $\frac{n+1}{2} - i$ with $j-1$ in the series of our recurrence relation, we get

$$3^{\frac{n+1}{2}} \times 2^M = 2^{M+n+1} - \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{M+n+1-2\times j}. \tag{21}$$

This result can also be obtained by using the remarkable identity $3^{\frac{n+1}{2}} - 4^{\frac{n+1}{2}}$.

We remark that this equation can be equivalent to the fifth form of the numbers which respect the Collatz conjecture and which have the form

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right). \tag{22}$$

And in this case we have the equation

$$2^{M+n+1} = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} \tag{23}$$

and we have also the equation

$$2^{M+n+1-2\times j} = 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \quad \text{for any natural number } j \text{ with } 1 \leq j \leq \frac{n+1}{2}. \tag{24}$$

However, in this case we should respect the fact that the numbers p_i can only be natural numbers which respect $p_i \geq 1 \; \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times N_0$, where N_0 can be any strictly positive natural number. Hence, since we should have $p_i \geq 1 \; \forall i \in \mathbb{N} \setminus \{0\}$, we should have

$$\sum_{i=j}^{\frac{n+1}{2}} p_i \geq \frac{n+1}{2} - j + 1 \tag{25}$$

And this means that we should have

$$2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \geq 2^{\frac{n+1}{2}-j+1} \quad \text{for any natural number } j \text{ with } 1 \leq j \leq \frac{n+1}{2} \tag{26}$$

Hence we should have

$$2^{M+n+1-2\times j} \geq 2^{\frac{n+1}{2}-j+1} \quad \text{for any natural number } j \text{ with } 1 \leq j \leq \frac{n+1}{2}. \tag{27}$$

We have

$$M + n + 1 - 2 \times j - \left(\frac{n+1}{2} - j + 1 \right) = M + \frac{n+1}{2} - 1 - j \tag{28}$$

And

$$\text{for any natural number } j \text{ with } 1 \leq j \leq \frac{n+1}{2}, \tag{29}$$

we have

$$M + \frac{n+1}{2} - 1 - j \geq M - 1 \geq 0 \quad \text{since } M > 0. \tag{30}$$

Consequently, we have always

$$2^{M+n+1-2\times j} \geq 2^{\frac{n+1}{2}-j+1}, \tag{31}$$

and this means that we respect indeed that, for any natural number j with $1 \leq j \leq \frac{n+1}{2}$, we have

$$2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \geq 2^{\frac{n+1}{2}-j+1} \tag{32}$$

We can remark also that, for $j = 1$, we have

$$p_0 = M + n + 1 - \sum_{i=j}^{\frac{n+1}{2}} p_i = M + n + 1 - (M + n + 1 - 2 \times j) = 2 \tag{33}$$

Consequently we respected also that p_0 is an even number. We can remark also that for $j = \frac{n+1}{2}$ we have

$$p_{\frac{n+1}{2}} = M + n + 1 - 2 \times j = M \tag{34}$$

Finally, we conclude that all the natural numbers which respect the first form of the numbers which respect the Collatz conjecture and which have the form $x_n = 2^M \ \forall M \in \mathbb{N} \setminus \{0\}$ respect also the fifth form

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right),$$

where n is odd and $n > 2$.

However, this result does not mean that $x_n = 2^M$ should be transformed to the number $x_0 = \frac{2^{2N}-1}{3}$ of the Collatz conjecture process after exactly n operations, as explained in [3], since we should only divide $x_n = 2^M$ by a power of two in order to reach the number 1 of the Collatz conjecture. Furthermore, in this case the representation of the natural number x_n with the fifth form is not unique, since the natural number n can take any odd value with $n > 2$.

3 About the uniqueness of the expressions of the fourth and fifth forms

Let us consider that we have two different expressions of the fifth form for a same number x_n which is a number of the fifth form. Then we have

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right)$$

and also

$$3^{\frac{m+1}{2}} \times x_n = 2^{\sum_{k=0}^{\frac{m+1}{2}} q_k} - \sum_{j=1}^{\frac{m+1}{2}} \left(3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \right),$$

where m is another odd natural number with $m > n$ and the numbers q_k are natural numbers which respect $q_k \geq 1 \ \forall k \in \mathbb{N} \setminus \{0\}$ and $q_0 = 2 \times M'$, where M' is a strictly positive natural number.

In this case we have two different "procedures" of the Collatz conjecture for the same even natural number x_n. Hence, by using the two equations, we can divide x_n by $2^{p_{\frac{n+1}{2}}}$ in order to get an odd number as a result and we can also divide x_n by $2^{q_{\frac{m+1}{2}}}$ in order to get an odd number as a result. This means obviously that

$$p_{\frac{n+1}{2}} = q_{\frac{m+1}{2}} \tag{35}$$

since the even natural number x_n is the same in the two equations, and we get these two equations as a result after our division:

$$\text{equation 1:}\quad 3^{\frac{n+1}{2}-1}\times\left(\frac{3\times x_n}{2^{p_{\frac{n+1}{2}}}}+1\right)=2^{\sum_{i=0}^{\frac{n+1}{2}-1}p_i}-\sum_{j=1}^{\frac{n+1}{2}-1}\left(3^{j-1}\times 2^{\sum_{i=j}^{\frac{n+1}{2}-1}p_i}\right) \tag{36}$$

$$\text{equation 2:}\quad 3^{\frac{m+1}{2}-1}\times\left(\frac{3\times x_n}{2^{q_{\frac{m+1}{2}}}}+1\right)=2^{\sum_{k=0}^{\frac{m+1}{2}-1}q_k}-\sum_{j=1}^{\frac{m+1}{2}-1}\left(3^{j-1}\times 2^{\sum_{k=j}^{\frac{m+1}{2}-1}q_k}\right). \tag{37}$$

Now by using these two equations, we can divide again $\frac{3\times x_n}{2^{p_{\frac{n+1}{2}}}}+1$ by $2^{p_{\frac{n+1}{2}-1}}$ in order to get an odd number as a result and we can also divide $\frac{3\times x_n}{2^{q_{\frac{m+1}{2}}}}+1$ by $2^{q_{\frac{m+1}{2}-1}}$ in order to get an odd number as a result. However, we proved above that

$$\frac{3\times x_n}{2^{p_{\frac{n+1}{2}}}}+1=\frac{3\times x_n}{2^{q_{\frac{m+1}{2}}}}+1. \tag{38}$$

And thus, this means that we have also

$$p_{\frac{n+1}{2}-1}=q_{\frac{m+1}{2}-1}. \tag{39}$$

We can carry on with the same steps until we prove that

$$p_{\frac{n+1}{2}-A}=q_{\frac{m+1}{2}-A}\quad\text{for any natural number } A \text{ with } 0\leq A\leq\frac{n+1}{2}. \tag{40}$$

Consequently, even if the natural number x_n that is a number of the fifth form has two different expressions of the fifth form, we remark in these different expressions that the numbers p_i and q_k (which are the exponents of the powers of two) are identical in both expressions starting from $p_0=q_{\frac{m-n}{2}}$ until $p_{\frac{n+1}{2}}=q_{\frac{m+1}{2}}$ (by varying the index A). Hence, we must calculate the numbers q_k separately for $0\leq k\leq\frac{m-n}{2}-1$.

We have

$$3^{\frac{m+1}{2}}\times x_n=3^{\frac{m-n}{2}}\times 2^{\sum_{i=0}^{\frac{n+1}{2}}p_i}-3^{\frac{m-n}{2}}\times\sum_{j=1}^{\frac{n+1}{2}}\left(3^{j-1}\times 2^{\sum_{i=j}^{\frac{n+1}{2}}p_i}\right) \tag{41}$$

Hence

$$3^{\frac{m+1}{2}}\times x_n=3^{\frac{m-n}{2}}\times 2^{\sum_{i=0}^{\frac{n+1}{2}}p_i}-\sum_{j=1}^{\frac{n+1}{2}}\left(3^{\frac{m-n}{2}+j-1}\times 2^{\sum_{i=j}^{\frac{n+1}{2}}p_i}\right) \tag{42}$$

Consequently

$$3^{\frac{m+1}{2}}\times x_n=3^{\frac{m-n}{2}}\times 2^{\sum_{i=0}^{\frac{n+1}{2}}p_i}-\sum_{j=\frac{m-n}{2}+1}^{\frac{m+1}{2}}\left(3^{j-1}\times 2^{\sum_{i=j}^{\frac{m+1}{2}}p_{i-\frac{m-n}{2}}}\right) \tag{43}$$

And we have

$$3^{\frac{m+1}{2}}\times x_n=2^{\sum_{k=0}^{\frac{m+1}{2}}q_k}-\sum_{j=1}^{\frac{m-n}{2}}\left(3^{j-1}\times 2^{\sum_{k=j}^{\frac{m+1}{2}}q_k}\right)-\sum_{j=\frac{m-n}{2}+1}^{\frac{m+1}{2}}\left(3^{j-1}\times 2^{\sum_{k=j}^{\frac{m+1}{2}}q_k}\right) \tag{44}$$

And since we proved that

$$p_{k-\frac{m-n}{2}}=q_k\quad\text{for any natural number } k \text{ with } \frac{m-n}{2}+1\leq k\leq\frac{m+1}{2} \tag{45}$$

then we get the equation

$$3^{\frac{m-n}{2}} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} = 2^{\sum_{k=0}^{\frac{m+1}{2}} q_k} - \sum_{j=1}^{\frac{m-n}{2}} \left(3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \right) \tag{46}$$

Consequently, we have

$$3^{\frac{m-n}{2}} = 2^{\sum_{k=0}^{\frac{m-n}{2}-1} q_k} - \sum_{j=1}^{\frac{m-n}{2}} \left(3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k - \sum_{i=0}^{\frac{n+1}{2}} p_i} \right) \tag{47}$$

However, we proved above also that

$$3^{\frac{m-n}{2}} = 2^{m-n} - \sum_{j=1}^{\frac{m-n}{2}} 3^{j-1} \times 2^{m-n-2\times j} \tag{48}$$

And thus, we get

$$2^{\sum_{k=0}^{\frac{m-n}{2}-1} q_k} = 2^{m-n} \quad \text{and} \quad 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k - \sum_{i=0}^{\frac{n+1}{2}} p_i} = 2^{m-n-2\times j} \quad \text{for } 1 \leq j \leq \frac{m-n}{2} \tag{49}$$

And since we have

$$\sum_{k=j}^{\frac{m+1}{2}} q_k - \sum_{i=0}^{\frac{n+1}{2}} p_i = \sum_{k=j}^{\frac{m-n}{2}-1} q_k \tag{50}$$

this means that

$$q_k = 2 \quad \text{for any natural number } k \text{ with } 0 \leq k \leq \frac{m-n}{2} - 1 \tag{51}$$

Finally, we conclude that, if the natural number x_n that is a number of the fifth form has two different expressions of the fifth form

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right)$$

and also

$$3^{\frac{m+1}{2}} \times x_n = 2^{\sum_{k=0}^{\frac{m+1}{2}} q_k} - \sum_{j=1}^{\frac{m+1}{2}} \left(3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \right),$$

where m is another odd natural number with $m > n$, then we remark in these different expressions that the numbers p_i and q_k (which are the exponents of the powers of two) are identical in both expressions for $p_{k-\frac{m-n}{2}} = q_k$ for any natural number k with $\frac{m-n}{2} \leq k \leq \frac{m+1}{2}$. And we have also $q_k = 2$ for $0 \leq k \leq \frac{m-n}{2} - 1$.

This result means not only that two different expressions of the fifth form are possible for the same natural number x_n, but it means also that, in order to find a second expression of the fifth form for a number x_n that has an expression of the fifth form

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right),$$

we can follow the following process, by considering an odd natural number m with $m > n$. We proved above this remarkable identity:

$$3^{\frac{m-n}{2}} = 2^{m-n} - \sum_{j=1}^{\frac{m-n}{2}} 3^{j-1} \times 2^{m-n-2\times j} \tag{52}$$

Hence, we have

$$3^{\frac{m-n}{2}} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} = 2^{m-n} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{m-n}{2}} 3^{j-1} \times 2^{m-n-2\times j} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} \tag{53}$$

We consider now that

$$\begin{aligned} &q_k = 2 \ \text{ for } 0 \leq k \leq \frac{m-n}{2} - 1 \quad \text{and} \\ &p_{k-\frac{m-n}{2}} = q_k \ \text{ for any natural number } k \text{ with } \frac{m-n}{2} \leq k \leq \frac{m+1}{2}. \end{aligned} \tag{54}$$

Consequently we have

$$3^{\frac{m-n}{2}} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} = 2^{\sum_{i=0}^{\frac{m+1}{2}} q_i} - \sum_{j=1}^{\frac{m-n}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \tag{55}$$

Hence

$$3^{\frac{m-n}{2}} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=\frac{m-n}{2}+1}^{\frac{m+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} = 2^{\sum_{i=0}^{\frac{m+1}{2}} q_i} - \sum_{j=1}^{\frac{m+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \tag{56}$$

And thus

$$3^{\frac{m-n}{2}} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - 3^{\frac{m-n}{2}} \times \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{n+1}{2}} q_{k+\frac{m-n}{2}}} = 2^{\sum_{i=0}^{\frac{m+1}{2}} q_i} - \sum_{j=1}^{\frac{m+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \tag{57}$$

Hence

$$3^{\frac{m-n}{2}} \times 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - 3^{\frac{m-n}{2}} \times \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{n+1}{2}} p_k} = 2^{\sum_{i=0}^{\frac{m+1}{2}} q_i} - \sum_{j=1}^{\frac{m+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \tag{58}$$

And thus, we have

$$3^{\frac{m-n}{2}} \times 3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{m+1}{2}} q_i} - \sum_{j=1}^{\frac{m+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \tag{59}$$

Finally, we conclude that

$$3^{\frac{m+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{m+1}{2}} q_i} - \sum_{j=1}^{\frac{m+1}{2}} 3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \tag{60}$$

And this means that the second expression of the fifth form for the natural number x_n is

$$3^{\frac{m+1}{2}} \times x_n = 2^{\sum_{k=0}^{\frac{m+1}{2}} q_k} - \sum_{j=1}^{\frac{m+1}{2}} \left(3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \right), \tag{61}$$

where m is another odd natural number with $m > n$ and the numbers q_k are natural numbers which respect $q_k \geq 1\ \forall k \in \mathbb{N} \setminus \{0\}$ and $q_0 = 2 \times M'$, where M' is a strictly positive natural number, since we have $q_k = 2$ for $0 \leq k \leq \frac{m-n}{2} - 1$ and $p_{k-\frac{m-n}{2}} = q_k$ for any natural number k with $\frac{m-n}{2} \leq k \leq \frac{m+1}{2}$.

We can conclude that, for an even number x_n that respects the expression of the fifth form, we can build infinitely many other different expressions of the fifth form for this same number x_n. Consequently, the fifth form expression of the number x_n is not unique, but we can choose the fifth form expression $3^{\frac{m+1}{2}} \times x_n = 2^{\sum_{k=0}^{\frac{m+1}{2}} q_k} - \sum_{j=1}^{\frac{m+1}{2}} \left(3^{j-1} \times 2^{\sum_{k=j}^{\frac{m+1}{2}} q_k} \right)$ where m is the smallest possible natural number.

And by making a similar proof, we can also make the same remark for any odd natural number x_n that is a number of the fourth form with two different expressions of the fourth form or more. We only have to start the proof by considering and using an even natural number

$$X_n = 3 \times x_n + 1 \tag{62}$$

since we have

$$3^{\frac{n}{2}+1} \times x_n + 3^{\frac{n}{2}} = 3^{\frac{n}{2}} \times (3 \times x_n + 1) = 3^{\frac{n}{2}} \times X_n. \tag{63}$$

4 The second form is included in the fourth form

We proved that

$$3^{\frac{n+1}{2}} \times 2^M = 2^{M+n+1} - \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{M+n+1-2\times j} = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right) \tag{64}$$

where $p_{\frac{n+1}{2}} = M$ and $p_0 = 2$, M can be any strictly positive natural number and n is an odd natural number. And thus we have also

$$3^{\frac{n+1}{2}} \times 2^{2\times M} = 2^{2\times M+n+1} - \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{2\times M+n+1-2\times j} = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right) \tag{65}$$

where

$$p_{\frac{n+1}{2}} = 2 \times M \quad \text{and} \quad p_0 = 2 \tag{66}$$

And we know that

$$3^{\frac{n+1}{2}} \times 2^{2\times M} = 3^{\frac{n+1}{2}} \times \left(2^{2\times M} - 1\right) + 3^{\frac{n+1}{2}} = 3^{\frac{n+1}{2}+1} \times \frac{\left(2^{2\times M} - 1\right)}{3} + 3^{\frac{n+1}{2}} \tag{67}$$

Hence we have

$$3^{\frac{n+1}{2}+1} \times \frac{\left(2^{2\times M} - 1\right)}{3} + 3^{\frac{n+1}{2}} = 2^{2\times M+n+1} - \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{2\times M+n+1-2\times j} \tag{68}$$

We have that $n+1$ is an even number. Let us consider that $m = n+1$, so that $m > 3$. We have

$$3^{\frac{m}{2}+1} \times \frac{\left(2^{2\times M} - 1\right)}{3} + 3^{\frac{m}{2}} = 2^{2\times M+m} - \sum_{j=1}^{\frac{m}{2}} 3^{j-1} \times 2^{2\times M+m-2\times j} \tag{69}$$

We remark that this equation can be equivalent to the fourth form of the numbers which respect the Collatz conjecture and which have the form

$$3^{\frac{m}{2}+1} \times x_m + 3^{\frac{m}{2}} = 2^{\sum_{i=0}^{\frac{m}{2}} p_i} - \sum_{j=1}^{\frac{m}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{m}{2}} p_i} \right) \tag{70}$$

And in this case we have the equation

$$2^{2\times M+m} = 2^{\sum_{i=0}^{\frac{m}{2}} p_i} \tag{71}$$

and we have also the equation

$$2^{2\times M+m-2\times j} = 2^{\sum_{i=j}^{\frac{m}{2}} p_i} \quad \text{for any natural number } j \text{ with } 1 \leq j \leq \frac{m}{2} \tag{72}$$

We have also

$$p_{\frac{m}{2}} = 2 \times M \quad \text{and} \quad p_0 = 2. \tag{73}$$

Like in the previous case, we should have, for any natural number j with $1 \leq j \leq \frac{m}{2}$,

$$2^{2\times M+m-2\times j} \geq 2^{\frac{m}{2}-j+1} \tag{74}$$

We have

$$2 \times M + m - 2 \times j - \left(\frac{m}{2} - j + 1 \right) = 2 \times M + \frac{m}{2} - 1 - j \tag{75}$$

And for any natural number j with $1 \leq j \leq \frac{m}{2}$, we have

$$2 \times M + \frac{m}{2} - 1 - j \geq 2 \times M - 1 \geq 0 \quad \text{since } M > 0. \tag{76}$$

Consequently, we have always

$$2^{2\times M+m-2\times j} \geq 2^{\frac{m}{2}-j+1}, \tag{77}$$

and this means that we respect indeed that $2^{\sum_{i=j}^{\frac{m}{2}} p_i} \geq 2^{\frac{m}{2}-j+1}$ for any natural number j with $1 \leq j \leq \frac{m}{2}$.

Finally, we conclude that all the natural numbers which respect the second form of the numbers which respect the Collatz conjecture and which have the form $x_m = \frac{2^{2\times M}-1}{3}$ $\forall M \in \mathbb{N} \setminus \{0\}$ respect also the fourth form

$$3^{\frac{m}{2}+1} \times x_m + 3^{\frac{m}{2}} = 2^{\sum_{i=0}^{\frac{m}{2}} p_i} - \sum_{j=1}^{\frac{m}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{m}{2}} p_i} \right)$$

with m even and $m > 3$.

However, this result does not mean that $x_m = \frac{2^{2\times M}-1}{3}$ should be transformed to the number $x_0 = \frac{2^{2N}-1}{3}$ of the Collatz conjecture process after exactly m operations, as explained in [3], since we have already $x_m = x_0 = \frac{2^{2\times M}-1}{3}$ without any operations related to the process of the Collatz conjecture. Furthermore, in this case the representation of the natural number x_m with the fourth form is not unique, since the natural number m can take any even value with $m > 3$.

5 The third form is included in the fifth form

We proved that

$$3^{\frac{m}{2}+1}\times\frac{\left(2^{2\times M}-1\right)}{3}+3^{\frac{m}{2}}=2^{2\times M+m}-\sum_{j=1}^{\frac{m}{2}}3^{j-1}\times 2^{2\times M+m-2\times j}=2^{\sum_{i=0}^{\frac{m}{2}}p_i}-\sum_{j=1}^{\frac{m}{2}}\left(3^{j-1}\times 2^{\sum_{i=j}^{\frac{m}{2}}p_i}\right) \tag{78}$$

where $p_{\frac{m}{2}}=2\times M$ and $p_0=2$, M can be any strictly positive natural number and $m=n+1$ is an even natural number with $m>3$. Let us consider that p is a strictly positive natural number. And thus, we have also

$$2^p\times\left(3^{\frac{m}{2}+1}\times\frac{\left(2^{2\times M}-1\right)}{3}+3^{\frac{m}{2}}\right)=2^p\times\left(2^{2\times M+m}-\sum_{j=1}^{\frac{m}{2}}3^{j-1}\times 2^{2\times M+m-2\times j}\right) \tag{79}$$

Consequently, we have

$$3^{\frac{m}{2}+1}\times\frac{\left(2^{2\times M+p}-2^p\right)}{3}=2^{2\times M+m+p}-3^{\frac{m}{2}}\times 2^p-\sum_{j=1}^{\frac{m}{2}}3^{j-1}\times 2^{2\times M+m+p-2\times j} \tag{80}$$

Let us consider that

$$\frac{m}{2}+1=\frac{L+1}{2}\ \Rightarrow\ L=m+1=n+2\quad\text{where } L \text{ is an odd natural number with } L>4. \tag{81}$$

We get

$$3^{\frac{L+1}{2}}\times\frac{\left(2^{2\times M+p}-2^p\right)}{3}=2^{2\times M+L+p-1}-3^{\frac{L+1}{2}-1}\times 2^p-\sum_{j=1}^{\frac{L+1}{2}-1}3^{j-1}\times 2^{2\times M+L+p-1-2\times j} \tag{82}$$

Let us consider that $p=p_{\frac{L+1}{2}}$, and since the number p is added to the powers of all the numbers $2^{2\times M+L+p-1-2\times j}$ for any natural number j with $1\leq j\leq\frac{L+1}{2}-1$, we remark that this equation can be equivalent to the fifth form of the numbers which respect the Collatz conjecture and which have the form

$$3^{\frac{L+1}{2}}\times x_L=2^{\sum_{i=0}^{\frac{L+1}{2}}p_i}-\sum_{j=1}^{\frac{L+1}{2}}\left(3^{j-1}\times 2^{\sum_{i=j}^{\frac{L+1}{2}}p_i}\right) \tag{83}$$

And in this case we have the equation $2^{2\times M+L+p-1}=2^{\sum_{i=0}^{\frac{L+1}{2}}p_i}$ and we have also these equations:

$$\begin{aligned}&2^{2\times M+L+p-1-2\times j}=2^{\sum_{i=j}^{\frac{L+1}{2}}p_i}\quad\text{for any natural number } j \text{ with } 1\leq j\leq\frac{L+1}{2}-1\\&\text{and}\quad 2^p=2^{p_{\frac{L+1}{2}}}.\end{aligned} \tag{84}$$

We can also remark in this case that $p_0=2$ and, like in the previous cases, we should have

$$2^{2\times M+L+p-1-2\times j}\geq 2^{\frac{L+1}{2}-j+1}\quad\text{for any natural number } j \text{ with } 1\leq j\leq\frac{L+1}{2}-1. \tag{85}$$

We have

$$2\times M+L+p-1-2\times j-\left(\frac{L+1}{2}-j+1\right)=2\times M+\frac{L}{2}+p-2-\frac{1}{2}-j \tag{86}$$

And for any natural number j with $1 \leq j \leq \frac{L+1}{2} - 1$, we have

$$2 \times M + \frac{L}{2} + p - 2 - \frac{1}{2} - j \geq 2 \times M + p - 2 \geq 0 \quad \text{since } M > 0 \text{ and } p > 0. \tag{87}$$

Consequently, we have always $2^{2\times M+L+p-1-2\times j} \geq 2^{\frac{L+1}{2}-j+1}$, and since we have $p > 0$ this means that we respect indeed that

$$2^{\sum_{i=j}^{\frac{L+1}{2}} p_i} \geq 2^{\frac{L+1}{2}-j+1} \quad \text{for any natural number } j \text{ with } 1 \leq j \leq \frac{L+1}{2}. \tag{88}$$

Finally, we conclude that all the natural numbers which respect the third form of the numbers which respect the Collatz conjecture and which have the form $3 \times x_L = 2^{2\times M+p} - 2^p \; \forall M \in \mathbb{N} \setminus \{0\}$ with $p > 0$ respect also the fifth form

$$3^{\frac{L+1}{2}} \times x_L = 2^{\sum_{i=0}^{\frac{L+1}{2}} p_i} - \sum_{j=1}^{\frac{L+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{L+1}{2}} p_i} \right)$$

with L odd and $L > 4$.

However, this result does not mean that $x_L = \frac{2^{2\times M+p}-2^p}{3}$ should be transformed to the number $x_0 = \frac{2^{2N}-1}{3}$ of the Collatz conjecture process after exactly L operations, as explained in [3], since only one operation of division by 2^p transforms x_L to $x_0 = \frac{2^{2N}-1}{3}$ in the process of the Collatz conjecture. Furthermore, in this case the representation of the natural number x_L with the fifth form is not unique, since the natural number L can take any odd value with $L > 4$.

6 Simple arithmetic statements equivalent to the proof of the Collatz conjecture

We could prove that all the strictly positive natural numbers which respect the Collatz conjecture can be expressed with only the following two simplified forms.

The first final form:

$$3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) \tag{89}$$

with N a natural number with $N \geq 2$.

And the second final form:

$$3^M \times x_{2M-1} = 2^{\sum_{i=0}^{M} p_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} p_i} \right) \tag{90}$$

with M a natural number with $M \geq 2$.

Here the numbers x_{2N} and x_{2M-1} are natural numbers which respect the Collatz conjecture and the numbers p_i can be any natural numbers which respect $p_i \geq 1 \; \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times A$, where A can be any strictly positive natural number.

We remarked that, for any natural number x_{2N} or x_{2M-1} that can be expressed in the first form, in the second form or in the third form described in the introduction and in [3], the expression of this natural number x_{2N} or x_{2M-1} is not unique in the fourth or in the fifth form, because we can use in that representation any natural number N or M with $N \geq 2$ and $M \geq 2$.

In order to prove the Collatz conjecture we should only prove that each strictly positive natural number can be expressed with one of these two simplified final forms. However, we can

remark that in the first final form x_{2N} should be an odd natural number, whereas in the second final form x_{2M-1} should be an even natural number. And thus, in order to prove the Collatz conjecture we should only prove that each strictly positive odd natural number can be expressed with the first final form and that each strictly positive even natural number can be expressed with the second final form.

We can remark that, if x is a strictly positive even natural number with $x \neq 2^M$, then $x = 2^p \times x_c$, where x_c is a strictly positive odd natural number and p is a strictly positive natural number. And if x_c can be expressed with the first final form, then we have

$$2^p \times \left(3^{N+1} \times x_c + 3^N\right) = 2^p \times \left(2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right)\right) \tag{91}$$

And this means that

$$3^{N+1} \times x = 2^p \times \left(2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right)\right) - 3^N \times 2^p \tag{92}$$

And this is equivalent to

$$3^{N+1} \times x = 2^{\sum_{i=0}^{N+1} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N+1} p_i}\right) - 3^N \times 2^p = 2^{\sum_{i=0}^{N+1} p_i} - \sum_{j=1}^{N+1} \left(3^{j-1} \times 2^{\sum_{i=j}^{N+1} p_i}\right) \tag{93}$$

where

$$p_{N+1} = p. \tag{94}$$

And since we proved that all the numbers of the form $x_n = 2^M \ \forall M \in \mathbb{N} \setminus \{0\}$ can be expressed with the second final form, we conclude finally that, in order to prove the Collatz conjecture, we should only prove that each strictly positive odd natural number can be expressed with the first final form

$$3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right)$$

with N a natural number with $N \geq 2$, where x_{2N} is the studied strictly positive odd natural number and the numbers p_i can be any natural numbers which respect $p_i \geq 1 \ \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times A$, where A can be any strictly positive natural number.

7 An infinite set of odd numbers respecting the first final form

Many may think that it is complicated to express any odd natural number with the first final form demonstrated in this article without the help of artificial intelligence or other mathematical software. However, there are many sequences and series of natural numbers which are equivalent to the set of the odd natural numbers and which can simplify this task. Furthermore, we can find many arithmetic developments of natural numbers which are easy recurrence relations. Let us give an example of a recurrence relation which develops a natural number into a series.

Let us consider a strictly positive natural number N. We study the number 4^N in order to deduce a useful result. For $N = 1$, $N = 2$ and $N = 3$ we have

$$4^1 = 3 \times 4^0 + 1 \tag{95}$$

$$4^2 = 3 \times 4^1 + 3 \times 4^0 + 1 \tag{96}$$

$$4^3 = 3 \times 4^2 + 3 \times 4^1 + 3 \times 4^0 + 1 \tag{97}$$

We can guess the recurrence relation

$$4^N = 1 + 3 \times \sum_{i=0}^{N-1} 4^i. \tag{98}$$

Now let us consider that this recurrence relation is correct up to the number N and let us prove it for $N+1$. We have

$$4^{N+1} = 4 \times 4^N = 4 + 3 \times \sum_{i=0}^{N-1} 4^{i+1} = 4 + 3 \times \sum_{i=1}^{N} 4^i = 4 - 3 + 3 \times \sum_{i=0}^{N} 4^i = 1 + 3 \times \sum_{i=0}^{N} 4^i \tag{99}$$

And thus, we conclude that

$$\frac{4^{N+1} - 1}{3} = \sum_{i=0}^{N} 4^i \tag{100}$$

And this means that our recurrence relation $4^N = 1 + 3 \times \sum_{i=0}^{N-1} 4^i$ is correct for any strictly positive natural number N.

This recurrence relation is an example that develops a number into a sum of other numbers, and it can be expressed also from the identity $A^N - 1 = (A-1) \times \sum_{i=0}^{N-1} A^i$. However, let us use it in the following proof.

Let us consider $y_B = \frac{2^{2B}-1}{3}$, where B is a strictly positive natural number. We have that y_B is a natural number satisfying the following formula:

$$3^{N+1} \times y_B + 3^N = 3^{N+1} \times \frac{2^{2B} - 1}{3} + 3^N = 3^N \times 2^{2B} \tag{101}$$

where N is a natural number with $N \geq 2$. And we proved above that

$$3^{\frac{n+1}{2}} \times 2^M = 2^{M+n+1} - \sum_{j=1}^{\frac{n+1}{2}} 3^{j-1} \times 2^{M+n+1-2\times j} \tag{102}$$

where n is an odd natural number with $n > 2$ and M is a strictly positive natural number. And thus, we have

$$3^{N+1} \times y_B + 3^N = 2^{2B+2N} - \sum_{j=1}^{N} 3^{j-1} \times 2^{2B+2N-2\times j} \tag{103}$$

where B and N are strictly positive natural numbers with $2B = M$ and $N = \frac{n+1}{2} \geq 2$. And we have

$$y_B = \frac{2^{2B} - 1}{3} = \frac{4^B - 1}{3} \quad \text{whereas we proved above that} \quad \frac{4^B - 1}{3} = \sum_{i=0}^{B-1} 4^i \tag{104}$$

And thus, the infinite set of odd numbers $y_B = \sum_{i=0}^{B-1} 4^i$, where B can be any strictly positive natural number, can all be expressed with the first final form

$$3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) \tag{105}$$

where $x_{2N} = y_B = \sum_{i=0}^{B-1} 4^i$ and N is a natural number with $N \geq 2$, by remarking that

$$\sum_{i=0}^{N} p_i = 2B + 2N \quad \text{where } p_0 = 2. \tag{106}$$

This demonstrated infinite set of odd numbers may serve as a basis in order to help the readers to prove that all the other odd natural numbers x_{2N} can be expressed with the first final form $3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right)$.

First corollary

We propose in this corollary a method to find an infinite amount of numbers which respect the Collatz conjecture by using the result above. Let us consider

$$y_C = C + \sum_{i=0}^{B-1} 4^i,$$

where C is a strictly positive natural number and B can be any strictly positive natural number. We have consequently

$$3^{N+1} \times y_C + 3^N = 3^{N+1} \times C + 2^{2B+2N} - \sum_{j=1}^{N} 3^{j-1} \times 2^{2B+2N-2\times j} \tag{107}$$

Let us consider that the new number y_C respects the Collatz conjecture by respecting the first final form with

$$3^{N+1} \times y_C + 3^N = 2^{2B+2N+2D} - \sum_{j=1}^{N} 3^{j-1} \times 2^{2B+2N-2\times j} \tag{108}$$

where D is a strictly positive natural number. We have in this case

$$\sum_{i=0}^{N} p_i = 2B + 2N + 2D \quad \text{where } p_0 = 2 + 2D. \tag{109}$$

And we have

$$2^{2B+2N+2D} = 3^{N+1} \times C + 2^{2B+2N} \tag{110}$$

And thus, we have

$$2^{2B+2N} \times \left(4^D - 1\right) = 3^{N+1} \times C \tag{111}$$

Let us consider that

$$C = E \times 2^{2B+2N} \quad \text{where } E \text{ is a natural number.} \tag{112}$$

Hence we get the equation

$$4^D = 3^{N+1} \times E + 1. \tag{113}$$

Now let us try to find examples of values for the natural numbers N, E and D which respect this equation. We can guess a first easy example of values, which is $N = 1$, $E = 7$ and $D = 3$. However, the number N is only interesting for us if $N \geq 2$. Hence, we guess a second interesting example of values:

$$N = 2 \ \text{ and } \ E = 9709 \ \text{ and } \ D = 9. \tag{114}$$

In this case, we have

$$C = E \times 2^{2B+2N} = 9709 \times 2^{2B+4} = 155344 \times 4^B \tag{115}$$

Now let us verify if the new number y_C really respects the first final form of the Collatz conjecture, where we have

$$y_C = C + \sum_{i=0}^{B-1} 4^i = 155343 \times 4^B + \sum_{i=0}^{B} 4^i. \tag{116}$$

We have

$$3^{N+1}\times y_C + 3^N = 3^3 \times \left(155343 \times 4^B + \sum_{i=0}^{B} 4^i\right) + 3^2 = 3^3 \times 155343 \times 4^B + 3^2 \times 4^{B+1} = 4194297 \times 4^B \tag{117}$$

And we have

$$2^{2B+2N+2D}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j}=2^{22}\times 4^{B}-4^{B+1}-3\times 4^{B}$$
$$=\left(2^{22}-4-3\right)\times 4^{B}=4194297\times 4^{B} \tag{118}$$

And thus, we conclude that the set of all the numbers $y_C=155343\times 4^B+\sum_{i=0}^{B}4^i$, where B can be any strictly positive natural number, respect the Collatz conjecture by respecting the first final form

$$3^{N+1}\times y_C+3^N=2^{2B+2N+2D}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j}$$

where $N=2$ and $D=9$.

Let us prove also that $y_C=155343\times 4^B+\sum_{i=0}^{B}4^i$ can never be expressed by the second form of the numbers that respect the Collatz conjecture, which is $\frac{2^{2\times M}-1}{3}$. In fact, we have

$$y_C\times 3+1=3\times 155343\times 4^B+4^{B+1}=466033\times 4^B=466033\times 2^{2B}\neq 2^{2\times M} \tag{119}$$

And thus, we have always $y_C\neq\frac{2^{2\times M}-1}{3}$. Consequently, the number y_C can never be expressed by the second form of the numbers that respect the Collatz conjecture. And thus, the demonstrated expression of the first final form is correct for a specific value of N, namely $N=2$, when we express each number of the set of numbers $y_C=155343\times 4^B+\sum_{i=0}^{B}4^i$ in the first final form.

We should remind that we did not consider that all the odd numbers should respect the formula

$$3^{N+1}\times y_C+3^N=2^{2B+2N+2D}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j}$$

because in the first final form of the Collatz conjecture we may have

$$\sum_{i=j}^{N}p_i\neq 2B+2N-2\times j. \tag{120}$$

However, the readers may find other examples of values for N, E and D for different other sets of numbers y_C, which are sets of numbers that change with the natural number B and which respect the Collatz conjecture without the change of the value of N. Another more general study is described in the following corollary.

Second corollary

We consider, like in the first corollary above, a new strictly positive natural number Y_{CN} with

$$3^{N+1}\times Y_{CN}+3^N=3^{N+1}\times C+2^{2B+2N}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j} \tag{121}$$

Let us consider again that Y_{CN} respects the Collatz conjecture by respecting the first final form with

$$3^{N+1}\times Y_{CN}+3^N=2^{2B+2N+2D}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j} \tag{122}$$

Like in the first corollary above, we have in this case $\sum_{i=0}^{N}p_i=2B+2N+2D$ where $p_0=2+2D$. And we have

$$2^{2B+2N+2D}=3^{N+1}\times C+2^{2B+2N} \tag{123}$$

And thus, we have

$$2^{2B+2N} \times \left(4^D - 1\right) = 3^{N+1} \times C \tag{124}$$

Let us consider again that

$$C = E \times 2^{2B+2N} \quad \text{where } E \text{ is a natural number.} \tag{125}$$

Hence we get the equation

$$4^D = 3^{N+1} \times E + 1. \tag{126}$$

Let us consider that $D = 3^N$ and we get the equation

$$E = \frac{4^{3^N} - 1}{3^{N+1}}. \tag{127}$$

This equation will be very useful if E is always a natural number. Let us investigate this equation by using the Fermat–Euler theorem. We have that the number 2 is coprime to the number 3^{N+1}, where N can be any strictly positive natural number with $N \geq 2$. And the Euler totient function of 3^{N+1} is

$$\phi\left(3^{N+1}\right) = 3^{N+1}\left(1 - \frac{1}{3}\right) = 2 \times 3^N. \tag{128}$$

Hence, we have by the theorem

$$2^{\phi\left(3^{N+1}\right)} \equiv 1 \bmod 3^{N+1}. \tag{129}$$

And thus, we proved that

$$4^{3^N} \equiv 1 \bmod 3^{N+1}. \tag{130}$$

We conclude that

$$E = \frac{4^{3^N} - 1}{3^{N+1}} \quad \text{is always a natural number.} \tag{131}$$

And this means that $E = \frac{4^{3^N}-1}{3^{N+1}}$ and $D = 3^N$ are always solutions of the equation of this corollary, which is $4^D = 3^{N+1} \times E + 1$, where N can be any strictly positive natural number with $N \geq 2$. We have generally in this case

$$C = E \times 2^{2B+2N} = \frac{4^{3^N} - 1}{3^{N+1}} \times 2^{2B+2N}. \tag{132}$$

We consider, like in the first corollary above, that

$$Y_{CN} = C + \sum_{i=0}^{B-1} 4^i = \frac{4^{3^N} - 1}{3^{N+1}} \times 2^{2B+2N} + \sum_{i=0}^{B-1} 4^i \tag{133}$$

Now let us verify if the new number Y_{CN} really respects the first final form of the Collatz conjecture. We have

$$\begin{aligned} 3^{N+1} \times Y_{CN} + 3^N &= 3^{N+1} \times \left(\frac{4^{3^N} - 1}{3^{N+1}} \times 2^{2B+2N} + \sum_{i=0}^{B-1} 4^i\right) + 3^N \\ &= \left(4^{3^N} - 1\right) \times 2^{2B+2N} + 3^{N+1} \times \sum_{i=0}^{B-1} 4^i + 3^N \end{aligned} \tag{134}$$

And we know that

$$3^{N+1} \times \sum_{i=0}^{B-1} 4^i = 3^N \times \left(4^B - 1\right) = 3^N \times 4^B - 3^N \tag{135}$$

Hence, we have

$$3^{N+1}\times Y_{CN}+3^N=\left(4^{3^N}-1\right)\times 2^{2B+2N}+3^N\times 4^B=4^B\times\left(4^{3^N+N}-4^N+3^N\right) \tag{136}$$

And we have

$$\begin{aligned}2^{2B+2N+2D}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j}&=4^B\times\left(4^{3^N+N}-\sum_{j=1}^{N}3^{j-1}\times 4^{N-j}\right)\\&=4^B\times\left(4^{3^N+N}-\sum_{j=0}^{N-1}3^{j}\times 4^{N-j-1}\right)\end{aligned} \tag{137}$$

And we know that

$$\sum_{j=0}^{N-1}3^j\times 4^{N-j-1}=4^N-3^N \tag{138}$$

Hence, we have

$$2^{2B+2N+2D}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j}=4^B\times\left(4^{3^N+N}-4^N+3^N\right) \tag{139}$$

And thus, we conclude that the set of all the numbers $Y_{CN}=\frac{4^{3^N}-1}{3^{N+1}}\times 2^{2B+2N}+\sum_{i=0}^{B-1}4^i$, where B can be any strictly positive natural number, respect the Collatz conjecture by respecting the first final form

$$3^{N+1}\times Y_{CN}+3^N=2^{2B+2N+2\times 3^N}-\sum_{j=1}^{N}3^{j-1}\times 2^{2B+2N-2\times j}$$

where N can be any strictly positive number with $N\geq 2$.

Let us prove that $Y_{CN}=\frac{4^{3^N}-1}{3^{N+1}}\times 2^{2B+2N}+\sum_{i=0}^{B-1}4^i$ can never be expressed by the second form of the numbers that respect the Collatz conjecture, which is $\frac{2^{2\times M}-1}{3}$. In fact, we have

$$Y_{CN}\times 3+1=\frac{4^{3^N}-1}{3^N}\times 2^{2B+2N}+4^B=2^{2B}\times\left(\frac{4^{3^N}-1}{3^N}\times 2^{2N}+1\right)\neq 2^{2M} \tag{140}$$

$$\text{this is because the natural number } \left(\frac{4^{3^N}-1}{3^N}\times 2^{2N}+1\right) \text{ is an odd number.} \tag{141}$$

And thus, we have always $Y_{CN}\neq\frac{2^{2\times M}-1}{3}$. Consequently, each number of the set Y_{CN} can never be expressed by the second form of the numbers that respect the Collatz conjecture. And thus, when we express each number of the set of numbers $Y_{CN}=\frac{4^{3^N}-1}{3^{N+1}}\times 2^{2B+2N}+\sum_{i=0}^{B-1}4^i$ with the demonstrated expression of the first final form, we use a specific value of N that changes with the change of Y_{CN}.

8 A contradiction argument for the numbers $4\times B+1$ and $4\times B-1$

8.1 Setting

In order to prove the Collatz conjecture, we should only prove that each strictly positive odd natural number can be expressed with the first final form (Section 5)

$$3^{N+1}\times x_{2N}+3^N=2^{\sum_{i=0}^{N}p_i}-\sum_{j=1}^{N}\left(3^{j-1}\times 2^{\sum_{i=j}^{N}p_i}\right),$$

with N a natural number with $N \geq 2$, where x_{2N} is the studied strictly positive odd natural number and the numbers p_i can be any natural numbers which respect $p_i \geq 1 \ \forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times A$, where A can be any strictly positive natural number. By Section 3, the expression of the first final form of a number is not unique: if $3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right)$ then also $3^{M+1} \times x_{2N} + 3^M = 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i}\right)$ with $M > N$, where $q_k \geq 1$ for $k \in \mathbb{N} \setminus \{0\}$, $q_0 = 2 \times M'$ (M' a strictly positive natural number), $q_k = 2$ for $0 \leq k \leq M - N - 1$ and $p_{k-(M-N)} = q_k$ for $M - N \leq k \leq M$.

In this section we consider that the sets of numbers $4 \times B + 1$ and $4 \times B - 1$ both respect the Collatz conjecture, with

$$3^{N+1} \times (4 \times B + 1) + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right) \quad \text{and}$$

$$3^{M+1} \times (4 \times B - 1) + 3^M = 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i}\right).$$

This implies that, in the expressions of the first final form, if $q_{N-1} = 6$ then we have $p_N = 4$ and $p_{N-1} = 1$ with $p_i = q_i$ for $0 \leq i \leq N - 2$. In the argument below this fact is shown to imply a contradiction, which is taken to prove that the Collatz conjecture is not true for all strictly positive natural numbers.

8.2 The consideration of the approach and its useful result

Let us consider that

$$3^{N+1} \times (4 \times B + 1) + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right) \tag{142}$$

where $4 \times B + 1$ is a natural number that respects the Collatz conjecture and B is a natural number with $B > 0$. Let us consider also that

$$3^{M+1} \times (4 \times B - 1) + 3^M = 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i}\right) \tag{143}$$

where $2 \times B - 1$ respects the Collatz conjecture too. This implies that $q_N = 1$ in the expression of the first final form, because $3^{M+1} \times (4 \times B - 1) + 3^M$ is only divisible by 2 and not 4.

We should remember that

$$4 \times B + 1 = 2 \times (2 \times A) + 1 \quad \text{and} \quad 4 \times B - 1 = 2 \times (2 \times A - 1) + 1 \tag{144}$$

with A a strictly positive natural number with $A > 1$. Hence, the sets of numbers $4 \times B + 1$ and $4 \times B - 1$ are separate sets.

Since M and N are not unique, as explained above, we can also make sure that, in the two expressions of the first final form that we use in this section,

$$M = N \quad \text{with } N > 2 \tag{145}$$

We have also

$$3^{M+1} \times (4 \times B - 1) + 3^M = 3^M \times (12 \times B - 2) = 3^N \times (12 \times B + 4) - 2 \times 3^{N+1}. \tag{146}$$

And we have

$$\begin{aligned} 3^{N+1} \times (4 \times B + 1) + 3^N &= 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) \\ &= 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i} \right) + 2 \times 3^{N+1}. \end{aligned} \tag{147}$$

And since we have $q_N = 1$ and $M = N$, then we have

$$\begin{aligned} 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) = 2^{\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-1} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} q_i} \right) \\ + 2 \times 3^{N+1} - 2 \times 3^{N-1} \end{aligned} \tag{148}$$

And thus, we have

$$\begin{aligned} 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N-1} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) - 2^{p_N} \times 3^{N-1} \\ = 2^{\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-1} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} q_i} \right) + 16 \times 3^{N-1} \end{aligned} \tag{149}$$

Hence

$$\begin{aligned} 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N-1} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) - 2^{p_N} \times 3^{N-1} \\ = 2^{\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} q_i} \right) - 2^{q_{N-1}+q_N} \times 3^{N-2} + 16 \times 3^{N-1} \end{aligned} \tag{150}$$

We consider that $q_{N-1} \geq 4$. Consequently, by making a comparison of parity between the two sides of the equation above, dividing by the powers of two, we remark that

$$p_N = 4. \tag{151}$$

Hence, we get

$$\begin{aligned} 2^{-4+\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N-1} \left(3^{j-1} \times 2^{-4+\sum_{i=j}^{N} p_i} \right) \\ = 2^{-4+\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{-4+\sum_{i=j}^{N} q_i} \right) - 2^{-3+q_{N-1}} \times 3^{N-2} + 2 \times 3^{N-1} \end{aligned} \tag{152}$$

And thus

$$\begin{aligned} 2^{-4+\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{-4+\sum_{i=j}^{N} p_i} \right) - 2^{p_{N-1}} \times 3^{N-2} \\ = 2^{-4+\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{-4+\sum_{i=j}^{N} q_i} \right) + 2 \times \left(3 - 2^{-4+q_{N-1}} \right) \times 3^{N-2} \end{aligned} \tag{153}$$

However, if $q_{N-1} > 4$ then we have

$$p_{N-1} = 1 \tag{154}$$

and we get

$$\begin{aligned} & 2^{-5+\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N-3} \left(3^{j-1} \times 2^{-5+\sum_{i=j}^{N} p_i}\right) - 2^{p_{N-2}} \times 3^{N-3} \\ & = 2^{-5+\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{-5+\sum_{i=j}^{N} q_i}\right) + \left(4 - 2^{-4+q_{N-1}}\right) \times 3^{N-2} \end{aligned} \tag{155}$$

Hence, when the natural numbers $4 \times B + 1$ and $4 \times B - 1$ both respect the Collatz conjecture with $B > 0$, we have always $q_N = 1$, and if $q_{N-1} = 6$ then we have

$$p_N = 4 \ \text{ and } \ p_{N-1} = 1 \ \text{ with } \ p_i = q_i \ \text{ for } \ 0 \leq i \leq N-2. \tag{156}$$

8.3 The contradictions implied by the useful result of the consideration

Let us consider that the natural numbers of the forms $4 \times B + 1$ and $4 \times B - 1$ all respect the Collatz conjecture with $B > 0$, and let us consider that $q_{N-1} = 6$. We get consequently

$$\begin{aligned} 3^{N+1} \times (4 \times B + 1) + 3^N &= 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right) \\ &= 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right) - 3^{N-2} \times 2^5 - 3^{N-1} \times 2^4 \end{aligned} \tag{157}$$

This means that

$$\frac{3^{N+1} \times (4 \times B + 1) + 3^N}{2^5} = 2^{\sum_{i=0}^{N-2} p_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N-2} p_i}\right) - 3^{N-2} - \frac{3^{N-1}}{2} \tag{158}$$

And we have also

$$\begin{aligned} 3^{N+1} \times (4 \times B - 1) + 3^N &= 2^{\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} q_i}\right) \\ &= 2^{\sum_{i=0}^{N} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} q_i}\right) - 3^{N-2} \times 2^7 - 3^{N-1} \times 2^1 \end{aligned} \tag{159}$$

This means that

$$\frac{3^{N+1} \times (4 \times B - 1) + 3^N}{2^7} = 2^{\sum_{i=0}^{N-2} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N-2} q_i}\right) - 3^{N-2} - \frac{3^{N-1}}{2^6} \tag{160}$$

However, we have also

$$p_i = q_i \quad \text{for } 0 \leq i \leq N-2. \tag{161}$$

Hence

$$2^{\sum_{i=0}^{N-2} p_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N-2} p_i}\right) = 2^{\sum_{i=0}^{N-2} q_i} - \sum_{j=1}^{N-2} \left(3^{j-1} \times 2^{\sum_{i=j}^{N-2} q_i}\right) \tag{162}$$

We get consequently

$$\frac{3^{N+1} \times (4 \times B + 1) + 3^N}{2^5} - \frac{3^{N+1} \times (4 \times B - 1) + 3^N}{2^7} = \frac{3^{N-1}}{2^6} - \frac{3^{N-1}}{2} \tag{163}$$

We remark that

$$\frac{3^{N+1}\times(4\times B+1)+3^N}{2^5}-\frac{3^{N+1}\times(4\times B-1)+3^N}{2^7}>0 \tag{164}$$

but we have also

$$\frac{3^{N-1}}{2^6}-\frac{3^{N-1}}{2}<0. \tag{165}$$

We conclude that this is a contradiction. And thus, we have always

$$q_{N-1}\neq 6. \tag{166}$$

This means that, when the separate sets of numbers $4\times B+1$ and $4\times B-1$ with $B>0$ both respect the Collatz conjecture, we have always

$$\frac{3^{N+1}\times(4\times B-1)+3^N}{2^7}\neq 2^{\sum_{i=0}^{N-2}q_i}-\sum_{j=1}^{N-2}\left(3^{j-1}\times 2^{\sum_{i=j}^{N-2}q_i}\right)-3^{N-2}-\frac{3^{N-1}}{2^6} \tag{167}$$

Hence, we have

$$3^{N-1}\times\frac{9\times B-1}{2^5}+3^{N-2}\neq 2^{\sum_{i=0}^{N-2}q_i}-\sum_{j=1}^{N-2}\left(3^{j-1}\times 2^{\sum_{i=j}^{N-2}q_i}\right) \tag{168}$$

for any natural number N with $N>2$ and for any natural number q_i that respects $q_i\geq 1$ $\forall i\in\mathbb{N}\setminus\{0\}$ with $i\leq N-2$ and respects $q_0=2\times M'$, where M' is a strictly positive natural number.

Since this result is true for any natural number N with $N>4$, we consider that $L=N-2$ and we get

$$3^{L+1}\times\frac{9\times B-1}{2^5}+3^{L}\neq 2^{\sum_{i=0}^{L}q_i}-\sum_{j=1}^{L}\left(3^{j-1}\times 2^{\sum_{i=j}^{L}q_i}\right)$$

for any natural number L with $L>2$ and for any natural number q_i that respects $q_i\geq 1$ $\forall i\in\mathbb{N}\setminus\{0\}$ with $i\leq N-2$ and respects $q_0=2\times M'$, where M' is a strictly positive natural number. This means that the natural numbers of the form

$$\frac{9\times B-1}{2^5}\quad\text{do not respect the Collatz conjecture.} \tag{169}$$

We can guess that, if $B=25$, then we have

$$\frac{9\times B-1}{2^5}=7 \tag{170}$$

This implies that the natural number 7 does not respect the Collatz conjecture. However, we have $7=4\times C-1$ with $C=2$, and we considered that all the natural numbers of the form $4\times B-1$ respect the Collatz conjecture. And thus, we can conclude that, if $4\times B+1$ and $4\times B-1$ both respect the Collatz conjecture, then this fact leads to a contradiction.

And thus, we conclude that not all the odd natural numbers can respect the Collatz conjecture at the same time. We conclude finally that the Collatz conjecture is not true for all strictly positive natural numbers. □

9 Summary and conclusion

We started this work by proving, thanks to a recurrence relation, that all the natural numbers which respect the first form of the numbers which respect the Collatz conjecture, $x_n = 2^M$ $\forall M \in \mathbb{N} \setminus \{0\}$, respect also the fifth form

$$3^{\frac{n+1}{2}} \times x_n = 2^{\sum_{i=0}^{\frac{n+1}{2}} p_i} - \sum_{j=1}^{\frac{n+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{n+1}{2}} p_i} \right).$$

We deduced that all the natural numbers which respect the second form, $x_m = \frac{2^{2\times M}-1}{3}$ $\forall M \in \mathbb{N} \setminus \{0\}$, respect also the fourth form

$$3^{\frac{m}{2}+1} \times x_m + 3^{\frac{m}{2}} = 2^{\sum_{i=0}^{\frac{m}{2}} p_i} - \sum_{j=1}^{\frac{m}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{m}{2}} p_i} \right),$$

and that all the natural numbers which respect the third form, $3 \times x_L = 2^{2\times M+p} - 2^p$ $\forall M \in \mathbb{N} \setminus \{0\}$ with $p > 0$, respect also the fifth form $3^{\frac{L+1}{2}} \times x_L = 2^{\sum_{i=0}^{\frac{L+1}{2}} p_i} - \sum_{j=1}^{\frac{L+1}{2}} \left(3^{j-1} \times 2^{\sum_{i=j}^{\frac{L+1}{2}} p_i} \right)$.

We concluded that the proof of the Collatz conjecture is equivalent to proving that each strictly positive odd natural number can be expressed with the first final form

$$3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i} \right) \quad (N \geq 2),$$

and that each strictly positive even natural number can be expressed with the second final form

$$3^M \times x_{2M-1} = 2^{\sum_{i=0}^{M} p_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} p_i} \right) \quad (M \geq 2),$$

where the numbers x_{2N} and x_{2M-1} are natural numbers which respect the Collatz conjecture and the numbers p_i can be any natural numbers which respect $p_i \geq 1$ $\forall i \in \mathbb{N} \setminus \{0\}$ and $p_0 = 2 \times A$, with A any strictly positive natural number. However, we finally concluded that, in order to prove the Collatz conjecture, we should only prove that each strictly positive odd natural number can be expressed with the first final form.

We added two corollaries in which infinite sets of odd numbers respecting the first final form are proved:

- The first general set of numbers is $y_B = \sum_{i=0}^{B-1} 4^i$, where B can be any strictly positive natural number and N is not unique. Furthermore, as a first corollary, we proposed a method to deduce other related infinite sets of numbers respecting the Collatz conjecture by using specific values of the natural number N. The proposed set of odd numbers is $y_C = 155343 \times 4^B + \sum_{i=0}^{B} 4^i$, where B can be any strictly positive natural number and N is specific with $N = 2$.

- The study in the second corollary is more general: for each natural number $N \geq 2$ we obtain a set of numbers respecting the Collatz conjecture by using specific values of N. The proposed set of odd numbers is $Y_{CN} = \frac{4^{3^N}-1}{3^{N+1}} \times 2^{2B+2N} + \sum_{i=0}^{B-1} 4^i$, where B and N can be any strictly positive natural numbers with $N \geq 2$, and we have

$$3^{N+1} \times Y_{CN} + 3^N = 2^{2B+2N+2\times 3^N} - \sum_{j=1}^{N} 3^{j-1} \times 2^{2B+2N-2\times j}.$$

This formula proves that each natural number $N \geq 2$ has many corresponding odd natural numbers Y_{CN} that respect the Collatz conjecture. However, we remark that these odd numbers grow rapidly and this means that we may need big calculators in order to use this formula.

If we want to prove that each strictly positive odd natural number x_{2N} can be expressed with the first final form, we should remember that the natural number N is not a constant for all the numbers x_{2N} and is not unique for a given x_{2N}, but we can choose the smallest possible natural number N for each x_{2N}. In fact, we proved in Section 3 that if

$$3^{N+1} \times x_{2N} + 3^N = 2^{\sum_{i=0}^{N} p_i} - \sum_{j=1}^{N} \left(3^{j-1} \times 2^{\sum_{i=j}^{N} p_i}\right)$$

then we have also

$$3^{M+1} \times x_{2N} + 3^M = 2^{\sum_{i=0}^{M} q_i} - \sum_{j=1}^{M} \left(3^{j-1} \times 2^{\sum_{i=j}^{M} q_i}\right)$$

where M is another natural number with $M > N$, the numbers q_k respect $q_k \geq 1 \ \forall k \in \mathbb{N} \setminus \{0\}$ and $q_0 = 2 \times M'$ (M' a strictly positive natural number), with $q_k = 2$ for $0 \leq k \leq M - N - 1$ and $p_{k-(M-N)} = q_k$ for $M - N \leq k \leq M$. If N is the smallest natural number satisfying the first final form for x_{2N}, all the other expressions are with $M = N + O$, where O can be any strictly positive natural number. This means that infinitely many other expressions of the first final form can be built for the same number x_{2N}. Using a simple recurrence relation for the proof of the Collatz conjecture is therefore not an easy task. Furthermore, the set of numbers Y_{CN} of the second corollary shows that the natural numbers N do not always increase when the odd natural number x_{2N} increases, since B can be any strictly positive natural number.

Consequently, for the proof that each strictly positive odd natural number x_{2N} can be expressed with the first final form, we propose to use a sequence $N = Clz(2k+1)$, where k can be any strictly positive natural number, and we will have to prove that

$$3^{Clz_{2k+1}+1} \times (2 \times k + 1) + 3^{Clz_{2k+1}} = 2^{\sum_{i=0}^{Clz_{2k+1}} p_i} - \sum_{j=1}^{Clz_{2k+1}} \left(3^{j-1} \times 2^{\sum_{i=j}^{Clz_{2k+1}} p_i}\right)$$

where $Clz_{2k+1} = Clz(2k+1)$ is a sequence of natural numbers that depends only on the natural number k. We have always

$$Clz(2k+1) = N = Clz(Y_{CN}) = Clz\left(\frac{4^{3^N} - 1}{3^{N+1}} \times 2^{2B+2N} + \sum_{i=0}^{B-1} 4^i\right)$$

where N depends on each natural number k by using the sequence Clz, we have always $Clz_{2k+1} \geq 2$, and B can be any strictly positive natural number.

In the appendix we added a new method to calculate all the possible values of the natural numbers $x_n = \frac{2^{2 \times M} - 1}{3} \ \forall M \in \mathbb{N} \setminus \{0\}$; the proposed algorithm is useful without any divisions, just by using the demonstrated sequence.

Finally, in Section 8 we considered that the separate sets of numbers $4 \times B + 1$ and $4 \times B - 1$ with $B > 0$ both respect the Collatz conjecture, each with its first final form. This consideration implied that we have always $q_N = 1$, and that if $q_{N-1} = 6$ then $p_N = 4$ and $p_{N-1} = 1$ with $p_i = q_i$ for $0 \leq i \leq N-2$. We then showed that $q_{N-1} = 6$ leads to a contradiction, and the author concludes that the Collatz conjecture cannot be true for all strictly positive natural numbers. The author suggests that researchers should now focus on finding the first counterexample of the Collatz conjecture.

A A sequence related to the second form of the numbers which respect the Collatz conjecture

We proved in [3] a recurrence relation for natural numbers k_n that is related to the second form of the numbers that respect the Collatz conjecture, and especially to the operation $3 \times x_n + 1 = 2^p$ that is the last operation of the Collatz conjecture before the last p successive divisions by two. The number x_n in this case is an odd number and

$$\begin{aligned} &3 \times x_n + 1 = 2^p \text{ is the last operation of the Collatz conjecture for } x_n \\ &\text{before the last } p \text{ successive divisions by two.} \end{aligned} \tag{171}$$

Hence, we consider that

$$x_n = 2 \times k_0 + 1 \quad \text{where } k_0 \text{ is a natural number.} \tag{172}$$

We get consequently

$$2^p = 3 \times (2 \times k_0 + 1) + 1 = 6 \times k_0 + 4 \tag{173}$$

Hence, when $p \geq 1$ we have

$$2^{p-1} = 3 \times k_0 + 2 \tag{174}$$

This means that k_0 is an even natural number. And thus, we consider that

$$k_0 = 2 \times k_1 \quad \text{where } k_1 \text{ is a natural number.} \tag{175}$$

We get consequently

$$2^p = 6 \times k_0 + 4 = 12 \times k_1 + 4 \tag{176}$$

Hence, when $p \geq 2$ we have

$$2^{p-2} = 3 \times k_1 + 1 \tag{177}$$

This means that k_1 is an odd natural number. We keep substituting the odd and even numbers k_n by expressions of k_{n+1} and we remark that k_2 is an even natural number and k_3 is an odd natural number, etc., and we remark the following recurrence relation, which is easy to prove:

$$2^p = 3 \times 2^{n+1} \times k_n + 2^{n+2} \quad \text{if } n \text{ is an even natural number} \tag{178}$$

and

$$2^p = 3 \times 2^{n+1} \times k_n + 2^{n+1} \quad \text{if } n \text{ is an odd natural number.} \tag{179}$$

We know that if $p = 4$ then $k_3 = 0 \Rightarrow k_2 = 0 \Rightarrow k_1 = 1 \Rightarrow k_0 = 2$. And thus

$$x_n = 2 \times k_0 + 1 = 5. \tag{180}$$

And we can verify that

$$3 \times x_n + 1 = 16 = 2^4. \tag{181}$$

Consequently, we remark that from this recurrence relation we can conclude the following sequence of natural numbers $(k_n)_{n \in \mathbb{N}}$, which respects $3 \times k_0 = 2^{p-1} - 2$ and $2 \times k_{n+1} = k_n - V_n$, where $(V_n)_{n \in \mathbb{N}}$ is a sequence defined by $V_n = 0$ if n is even and by $V_n = 1$ if n is odd. Furthermore, this sequence of natural numbers respects the fact that $x_n = 2 \times k_0 + 1$.

And thus, we can use this sequence starting from $n = N$ where $k_N = 0$. We have in this case $N = p - 1$ and we can calculate the value of k_0 with the sequence in order to calculate the value of x_n. By using this sequence, we get an infinite set of values of the number x_n which respect the fact that $3 \times x_n + 1 = 2^p$. However, we remark that the natural number p is even for all the calculated values of the number x_n. This means that we should make the following arithmetic investigation about the natural number p. We have

$$2^p \bmod 3 = (2 - 3i)^p \bmod 3 \quad \text{for any natural number } i. \tag{182}$$

For $i = 1$, we have

$$2^p \bmod 3 = (-1)^p \bmod 3 \tag{183}$$

However, we have $3 \times x' = 2^p - 1$, where x' is an odd natural number, which means that we have

$$2^p \equiv 1 \bmod 3 \tag{184}$$

Consequently, we have always

$$(-1)^p \equiv 1 \bmod 3 \tag{185}$$

And this is only possible when p is an even number.

We conclude that the proposed sequence is a good method to calculate all the values of $\frac{2^{2\times M}-1}{3}$, where M is a strictly positive natural number, with only an easy summation by considering that $k_{p-1} = 0$ and without any divisions. This method can also be very useful for the study of the second form of the numbers that respect the Collatz conjecture.

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